

What CA2 contributes in this modeled circuit

Controlled input-pattern experiment | CARLsim on the local NVIDIA GPU

We asked whether CA2 pyramidal output makes two equally strong cortical input patterns easier to distinguish in a downstream CA1 sample, and whether competing CA3 input changes that contribution. This is a test of information transmission in a fitted model. It is not a simulation of social recognition, learning, or a particular animal.

Primary result: information in CA1 spikes

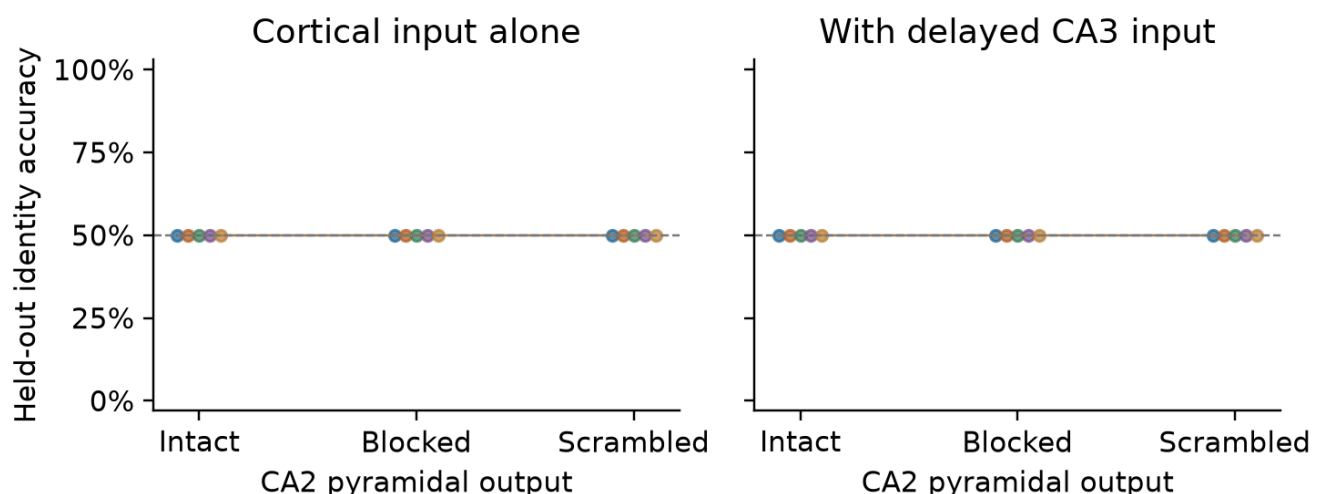
Across all five nominal networks, every condition scored 50% with the frozen four-window spike decoder. Cortical input alone produced no CA1 pyramidal spikes; adding CA3 produced five spikes per target in every trial. This gives an uninformative spike-count readout under the tested conditions, not evidence that CA2 lacks a biological function.

CA2 pyramidal output	Held-out spike accuracy	Rate-only accuracy
intact	50.0%	50.0%
blocked	50.0%	50.0%
scrambled	50.0%	50.0%

Intact minus blocked: +0.0 points; descriptive 95% network bootstrap interval +0.0 points to +0.0 points.

Intact minus scrambled: +0.0 points; descriptive 95% network bootstrap interval +0.0 points to +0.0 points.

Primary: identity information in CA1 spikes



Predetermined train/test split and nearest-centroid classifier. Reference centroids are learned separately for each network, context and output condition, using only designated training trials. Each point is a model network; chance is 50%. Five networks are not five animals.

This asks how much pattern information the activity makes available to the chosen readout after it is trained for that condition. It does not test an unchanged learned readout during an acute biological lesion. The analysis rule stays fixed; its training centroids differ between conditions.

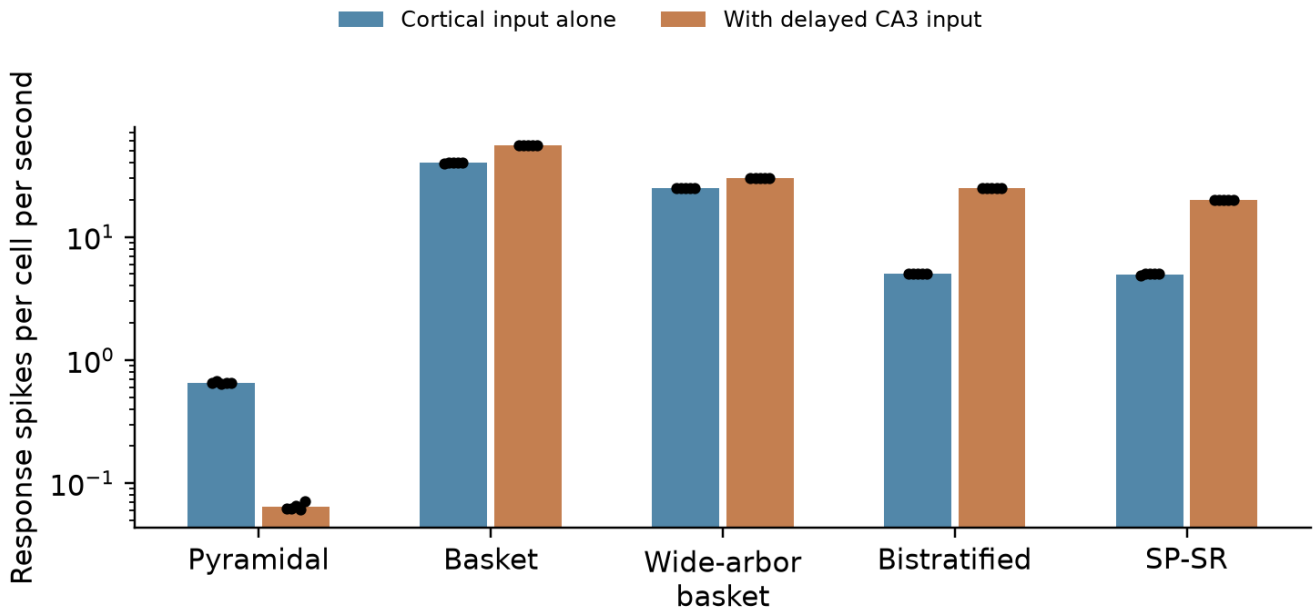
A null spike result limits this readout and these conditions. It does not show that CA2 has no function in an animal. Silent or saturated target responses, omitted pathways, and the fitted parameters can limit the test.

The zero-width bootstrap interval follows from all five measured spike effects being zero in this assay. It is not a statement that the biological effect is exactly zero, or that every possible spike-timing readout would fail.

What changed inside CA2?

Across the nominal trials, adding the CA3 input context coincided with a fall in mean CA2 pyramidal response from 2,475 to 244 spikes per trial, about 90%. The responsive pyramidal fraction fell from 13.1% to 1.2%. All four inhibitory populations increased their response rates. This is a clear context-dependent change inside the fitted model even though the chosen CA1 spike-count readout did not distinguish the input classes.

CA2 activity under the two input contexts



Bars show mean response rates across five networks; black points are network means. The vertical scale is logarithmic. Rates average spikes over the 200 ms response window and all cells in each modeled population; they are not sustained in-vivo firing rates.

This comparison is descriptive: cortical jitter is drawn separately in the two contexts, rather than held exactly identical. The inhibitory gain was calibrated to CA3 inhibition, so suppression is conditional on that fit and is not an independent discovery of a CA2 function. Core parameter and higher-precision checks in `context-response-results.json` test whether the pattern depends on those choices.

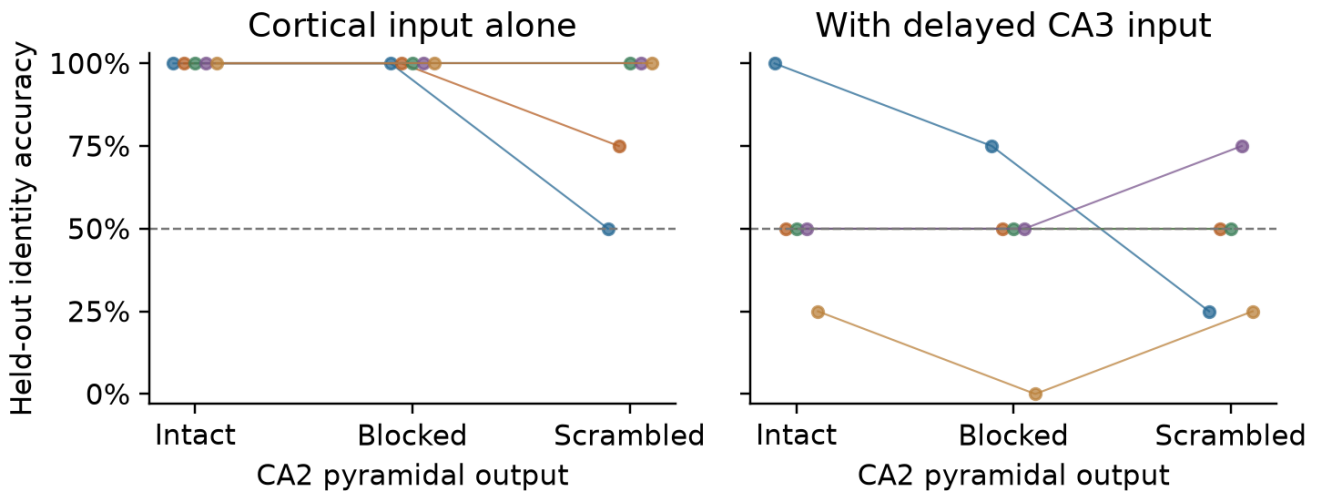
The direction depends on fitted inhibition: halving the inhibitory multiplier to 50 changes network 31 from suppression to an approximately 26% increase in CA2 pyramidal response with CA3 input. Suppression persists with the tested reversal changes, multiplier 150, and higher precision. This identifies inhibitory strength as a decisive model assumption; it does not establish which regime an animal occupies.

`Response-dynamics-results.json` also records population and responsive-cell latencies, responsive fractions, repeat similarity, and between-class separation for every modeled population. Silent vectors have undefined similarity and are retained as missing; silent cells are not assigned an artificial spike latency. These secondary descriptions do not replace the primary readout.

Secondary result: information in voltage responses

The prespecified secondary measure uses mean membrane voltage in four response windows. It can reveal differences below the threshold for generating spikes. It does not replace the primary result and is not called spike transmission or memory.

Secondary: identity information in CA1 voltage



Same input labels, training/test split, and classifier as the primary measure. Baseline-subtracted voltage features use the 128 modeled CA1 pyramidal targets.

Intact minus blocked: +5.0 points; descriptive 95% network bootstrap interval +0.0 points to +10.0 points.

Intact minus scrambled: +12.5 points; descriptive 95% network bootstrap interval -5.0 points to +37.5 points.

Interpret voltage differences only together with the higher-precision and parameter sensitivity results below. Small voltage information can be numerically fragile even when population spike counts change little.

With cortical input alone, voltage classification is 100% in both intact and pyramidal-blocked conditions in every network. The preserved CA2 inhibitory streams can still convey the input pattern. Thus successful voltage classification by itself does not establish a contribution of CA2 pyramidal output. The paired intervention differences, rather than raw accuracy, address that question.

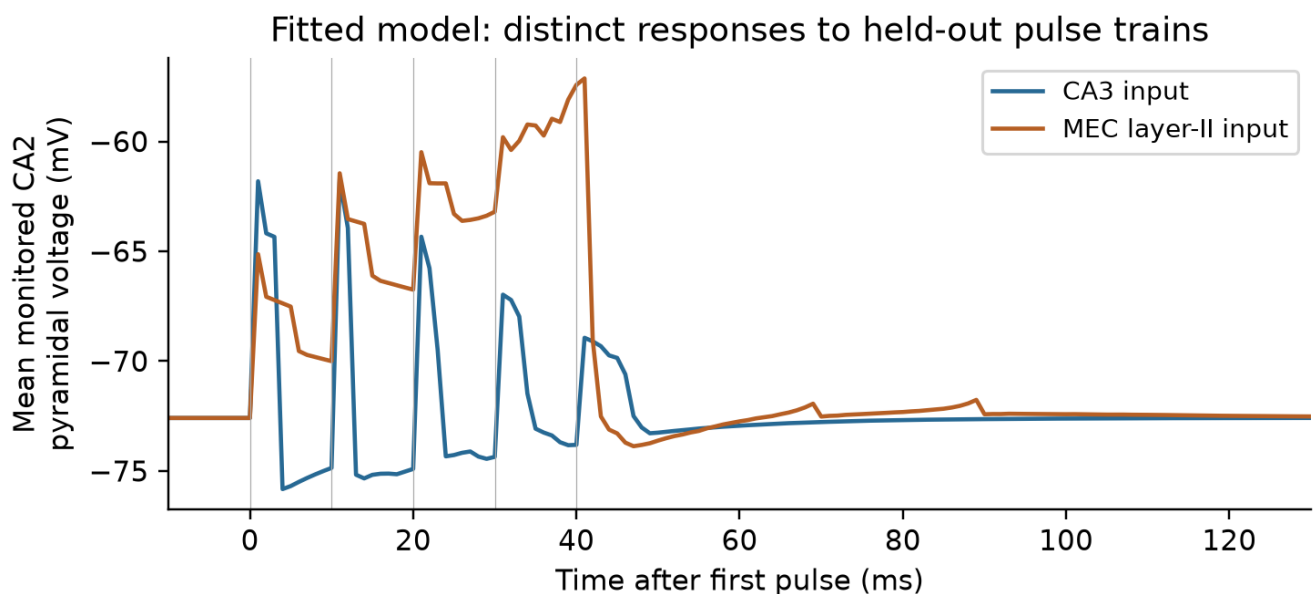
What the controls isolate

Blocking deletes only CA2 pyramidal output. All four CA2 inhibitory streams and the direct CA3 stream stay fixed. Scrambling changes pyramidal cell identities while preserving every spike time, population count, and cell train up to a bijective remapping. This tests whether cell-specific output contributes beyond the amount and timing of activity.

Why the unchanged export was insufficient

Matching measured ingredients does not guarantee that the assembled circuit reproduces a measured response. After matching single-pulse excitation, the unchanged model produced a CA3 inhibitory voltage difference of about 0.27 mV in magnitude; the published target was 13.8 $\pm$ 2.4 mV. We retained this negative result.

The revised model uses an assumed GABA_A reversal of -77.8 mV and a fitted multiplier of 100 on the four inhibitory-to-CA2-pyramidal connections. Other weights retain exported values. This large phenomenological correction is an important limitation, not a measured synaptic conductance. The reversal estimate comes from a recording recipe, not a direct GABA_A reversal measurement.



Unused 100 Hz pulse trains check qualitative pathway responses. Traces average the 128 monitored CA2 pyramidal cells, not every CA2 cell. Spike counts cover all modeled cells. Model observations are not animal replicates. All failed earlier checks remain archived.

The reference physiology is Chevaleyre and Siegelbaum (2010), [Strong CA2 pyramidal neuron synapses define a powerful disynaptic cortico-hippocampal loop](#). Calibration uses aggregate entorhinal layer-II and Schaffer collateral excitation and CA3 inhibition. Layer-III response targets are not silently applied to the modeled MEC layer-II source. Calibration is not independent biological validation.

Downstream biological validation remains limited

A separate CA1 check used the same 1,940-axon single CA3 pulse calibrated against CA2. All 128 modeled CA1 targets fired, both with inhibition intact and with target inhibition removed. This was not an eligible subthreshold EPSP comparison. The model also rested near -69.36 mV, whereas the reference cells were held at -73 mV, so the check is not a matched replication. No source parameter was changed to improve the nominal task result.

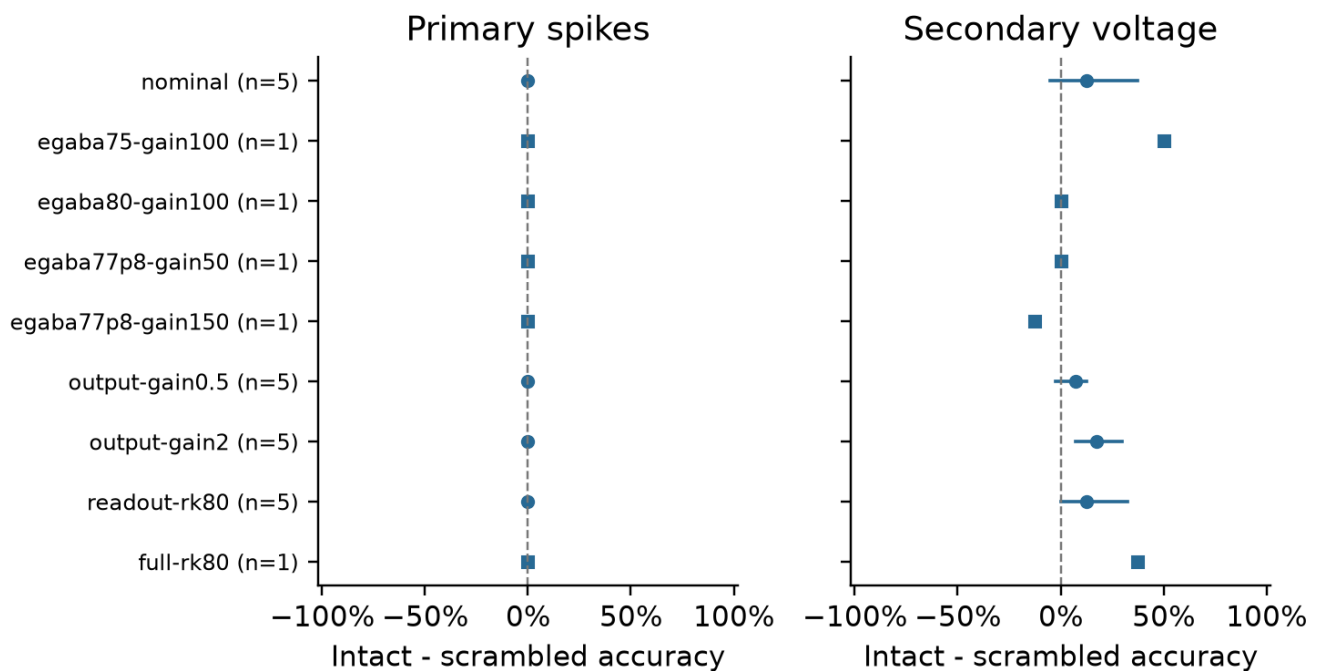
Thus numerical reliability does not establish biological validity of the downstream assay. Any null or voltage-based routing result must retain this limitation; strong direct CA3 drive can mask a smaller CA2 contribution.

Model boundaries

The core contains the five exported CA2 types at full counts. Inputs are prescribed MEC layer-II and CA3 events. The downstream assay has 128 CA1 pyramidal cells and selected full-count inhibitory populations. There is no upstream feedback, CA1 pyramidal feedback, explicit dendrite, spatial integration, or long-term learning. This model is not specifically ventral CA1.

Sensitivity: how dependent is the result?

Effect sensitivity: fixed analysis rules and matched inputs



Paired intact-minus-scrambled accuracy effects, using the same feature definitions, classifier algorithm and train/test split. Centroids are refitted from each variant and condition's training responses. Core perturbations and full-core precision use network 31; output-strength and readout-precision checks use all five nominal networks. Perturbation ranges are exploratory, not measured confidence bounds.

The sensitivity analysis changes the assumed inhibitory reversal (-75/-80 mV), the fitted inhibitory multiplier (50/150), and three default-flagged CA2 output connections (half/double effective strength). It also repeats readout calculations at 80 substeps/ms and repeats the complete core task at that precision for network 31.

Reversal variants affect GABA_A synapses globally in both CA2 and CA1. Their downstream effects therefore cannot be attributed solely to CA2 inhibition. Gain variants affect only inhibitory-to-CA2-pyramidal connections; output variants affect only the three represented default-flagged CA2 pyramidal output connections.

Additional 40/80-substep spot checks cover both contexts in four core variants: 7 of 8 meet the pre-existing tolerances. With gain 150 and no CA3 input, intact CA1 basket counts change from 39 to 36 (7.7%, above the 5% tolerance); this case remains numerically unresolved. Two preselected trials per variant cannot establish full classifier convergence. Exact checks are in variant-precision-results.json.

Any effect that changes direction or disappears across these checks is conditional or fragile. A convenient alternative is not selected to replace the frozen nominal result. The full JSON results include both blocked and scrambled comparisons, all per-network values, and the scope of every check.

Every tested variant retains chance-level primary spike classification. Secondary voltage effects depend on assumptions: the intact-minus-blocked effect changes sign in several variants. Doubling the default output strengths produces positive descriptive intervals, but does not validate those assumed strengths. At higher downstream precision the five-network intact-minus-scrambled mean remains +12.5 points; at full-core higher precision the network-31 effect falls from +62.5 to +37.5 points. These results support a conditional voltage signal rather than a robust biological function.

Does the fixed readout generalize?

We prospectively added unseen variants of the same two synthetic classes, replacing 10% or 50% of their distinctive cells while preserving total activity. This changes 1% or 5% of the full active ensemble. We also tested CA3 delays of 15 and 35 ms instead of 25 ms. Centroids stayed fixed to the nominal training trials; these new data were never used for training.

Unseen test	Intact	Blocked	Scrambled	CA2 pyramidal spikes
delay15, CA3 present	50.0% / 50.0%	50.0% / 50.0%	50.0% / 50.0%	281.0
delay35, CA3 present	50.0% / 50.0%	50.0% / 50.0%	50.0% / 50.0%	1,590.0
pattern10, CA3 absent	50.0% / 100.0%	50.0% / 95.0%	50.0% / 55.0%	2,476.3
pattern10, CA3 present	50.0% / 55.0%	50.0% / 50.0%	50.0% / 40.0%	244.8
pattern50, CA3 absent	50.0% / 55.0%	50.0% / 70.0%	50.0% / 50.0%	2,466.9
pattern50, CA3 present	50.0% / 35.0%	50.0% / 45.0%	50.0% / 45.0%	242.1

Each entry gives mean spike / voltage classification accuracy across five networks, with four new trials per case and network. Paired effects and descriptive network bootstrap intervals are recorded in `generalization-results.json`. Labels follow the majority of the original distinctive pool; the larger change intentionally makes the patterns close to balanced. All tested input patterns must remain distinguishable by the fixed input positive control. These tests do not establish generalization to animal behavior.

All unseen tests retain 50% spike accuracy. For the smaller pattern change without CA3, voltage accuracy is 100% intact, 95% blocked and 55% scrambled. The paired intact-minus-scrambled difference is +45 points (descriptive interval +35 to +50), but intact-minus-blocked is only +5 points (0 to +15): preserved inhibitory streams can still carry class information. The larger pattern change loses this consistent advantage. Both new CA3 delays give 50% voltage accuracy in every condition. Generalization is therefore limited to particular inputs and measures.

The final column gives mean upstream CA2 pyramidal response spikes per trial across all modeled cells. Other core response rates, responsive fractions and latencies are also retained. New trials use fresh jitter, so timing comparisons with nominal trials are descriptive rather than exact matched-jitter counterfactuals. These response measures do not tune or replace the primary decoder.

Uncertainty and scope

Five independently generated networks support descriptive uncertainty about this model. They do not sample animals, input states in natural behavior, or the uncertainty of every exported parameter. The single-network core sensitivity screen does not establish robustness across networks. Balanced label-permutation results are diagnostic and unadjusted for multiple comparisons.

Exact experiment and evidence

Item	Frozen nominal configuration
CA2 populations	18,956 pyramidal; 105 basket; 147 wide-arbor basket; 79 bistratified; 94 SP-SR
Prescribed input	10,818 MEC source cells; 9,800 active in each pattern; 8,820 shared. CA3 pool 4,096; 1,940 active when present.
Timing	Five 100 Hz events with 0-2 ms jitter. Cortical onset 50 ms; CA3 onset 75 ms. Trial 250 ms; response 50-249 ms.
State and replication	Fresh process for each trial; network seeds 31-35; 16 trials/network; calibration seed 20 excluded.
Readout	128 CA1 targets; 819 basket and 1,962 bistratified cells. Four 50 ms windows; two training and two test repetitions/pattern/context.
Numerics	RK4 40 substeps/ms; core checks 20/40, readout checks 40/80; direct checks of inferred effects at higher precision.
Synapses	Exported pair-specific short-term plasticity; AMPA/GABAa active; NMDA/GABAb multipliers zero; no long-term learning.
Audit	80 core trials and 240 matched replays verified.

The independent audit checks exact input delivery, intact/blocked/bijective-scrambled streams, unchanged inhibitory and CA3 streams, full records for monitored voltage subsets, spikes from all modeled cells, reset baselines, and aggregate connection-count signatures. The latter are not individual-edge hashes; fixed code and seeds support topology matching.

The extended audit additionally verifies 208 core runs and 1,344 matched readouts across parameter changes, higher precision and unseen-input tests. Together with the nominal experiment, this covers 288 core runs and 1,584 readouts. Numerical spot-check failures remain scientific limitations despite successful execution and data-integrity checks.

Files: frozen-task-protocol.json and nominal-parameter-classification.json record configuration and provenance; reset-task-results.json, secondary-voltage-results.json, sensitivity-results.json, and generalization-results.json contain analyses; nominal-experiment-audit.json records verification; methods.txt gives detailed assumptions and reproducibility instructions. Runs and replays contain exact event files, executable/input hashes, configurations, logs, and binary outputs.

The original frozen analysis script is retained. A documented variable-name collision was corrected in analyze_reset_task_v2.py before test-set analysis; the feature and classifier function definitions and inference rules are unchanged. The corrected ingestion was checked on completed real replays.

Backend: CARLsim4 branch feat/meansdSTPPost_hc, source commit 11ea96f750d125e4dcfbf67231429204322fa1bd, with the documented GPU architecture, monitor-capacity and inhibitory-reversal changes. This run uses the archived local implementation and frozen executable, rather than assuming equivalence to another simulator release.

The preserved source archive hash and embedded commit identifier were checked directly. backend-source-provenance.json records modified source/build-file hashes, library hashes, exact build logs and compiler warnings. Local changes also provide CUDA 12 compatibility and timer/check helpers. The archived builds retain CPU thread-wrapper return-type warnings; this study uses GPU mode and does not validate CPU simulation.

REPRODUCE.txt gives the full sequential rerun workflow. prepare_reproduction_copy.py creates a fresh workspace while retaining the exact historical unit-gain reference separately. An isolated copy was checked for all frozen hashes, nominal and unseen input files, CUDA library resolution and analysis/PDF prerequisites. This was a copy and dependency check, not a second full GPU rerun.

Hippocampome source: [simulation parameter export](#). The implemented rows include database defaults; an unflagged row is not proof that each field is directly measured. Effective fitted neuron parameters differ from a matched biological cell model.

The defensible conclusion concerns what this fitted circuit transmits under these controlled conditions, and which assumptions determine that result. Establishing a CA2 function in animals requires matched experimental recordings and biological validation beyond this assay.